

# KGML notebook

2025-03-12

## Conversions

solving for the conversion factor from plot yield to kilograms per hectare

$$y = mx$$

where  $y$  is grain yield in desired unit, likely kg ha

where  $x$  is grain yield in grams per harvest area

where  $m$  is the conversion factor

## Fert

### corn and soy yields

$$\begin{aligned} \frac{\text{grain yield (g)}}{\text{plot}} \times \frac{\text{plot}}{4.572 \text{ m}^2} \times \frac{10000 \text{ m}^2}{1 \text{ hectare}} \times \frac{1 \text{ kg}}{1000 \text{ g}} &= 2.187227 \\ \text{plot yield (g plot}^{-1}) \times 2.187 &= \text{plot yield (kg ha}^{-1}) \\ y &= 2.187 x \end{aligned}$$

### IWG yields

$$\begin{aligned} \frac{\text{iwg weight } g}{\text{sample area}} \times \frac{\text{sample area}}{1.672 \text{ m}^2} \times \frac{10000 \text{ m}^2}{1 \text{ hectare}} \times \frac{1 \text{ kg}}{1000 \text{ g}} &= 5.98 \\ y &= 5.98 x \end{aligned}$$

## Issues with data

### ORG

- outliers for corn and soybean yields, likely remove from data
- IWG yields have some values above 2000 kg ha

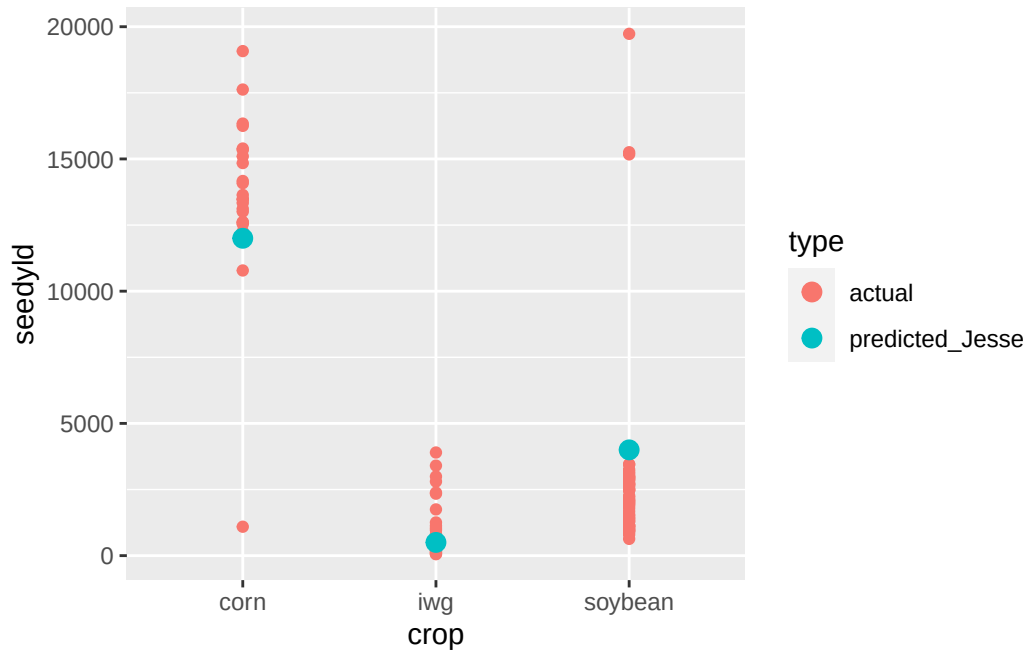
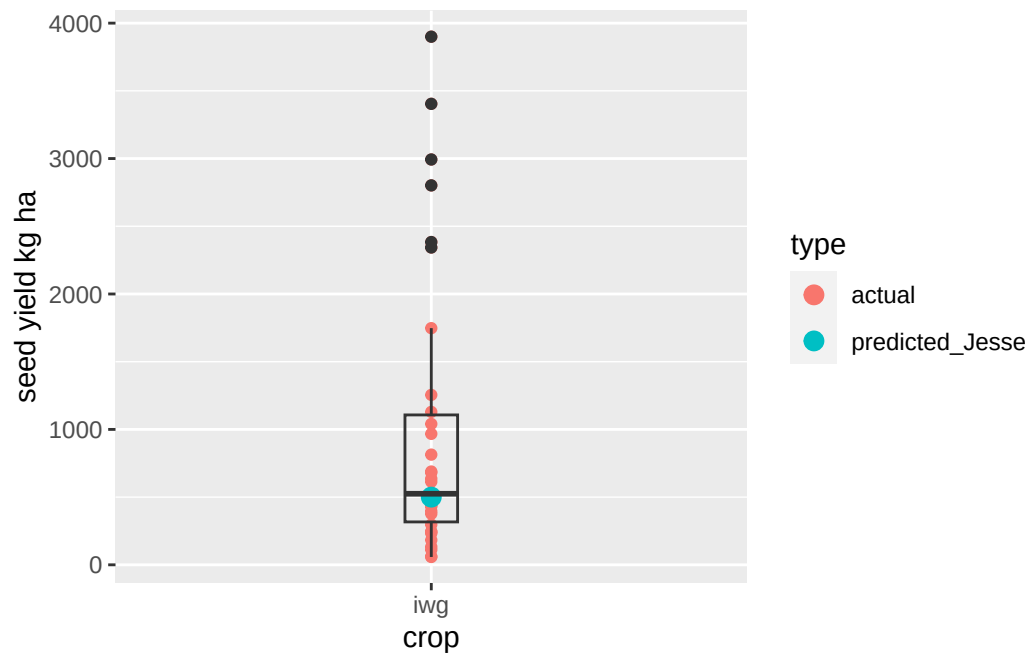


Table 1: estimated seed yield by crop in kg ha. I used a conversion from bushels acre yields + test weight to kg ha

| Crop       | Jesse            | ChatGPT          | Google           |
|------------|------------------|------------------|------------------|
|            | seed yield kg ha | seed yield kg ha | seed yield kg ha |
| corn       | 12000            | 7000-12000       | 7000             |
| soybean    | 4000             | 2000-45000       | 4000-11000       |
| wheatgrass | 500              | 600              | 600              |

### Closer look at IWG



- kernza grain yields above 2000 kg ha are suspicious, and are noted as a boxplot to be outliers